Lecture 3

Homotopy and the fundamental group

Objectives. *We study which loops can be continuously deformed to a constant loop. A loop around a missing point in the plane cannot be contracted without meeting that point. We prove this obstruction by following a continuous phase around the circle. We compare paths by deformations that keep their endpoints fixed. A path returning to its starting point is a loop. The classes of loops at one fixed point form the fundamental group. For the circle this group is the integers: a continuously chosen angle records how many turns a loop makes. We prove this by lifting paths and homotopies, then use the integer to prove the fundamental theorem of algebra.*

By the end of this lecture you will be able to:

- Construct the fundamental group, induced homomorphisms and base-point isomorphisms.
- Prove that lifting computes the fundamental group of the circle as $\mathbb{Z}$.
- Compute the group of the punctured plane and prove the disc has no retraction onto its circle.
- Prove that a nonconstant complex polynomial has a root.

Unit 3A Homotopies and continuous phases

3.1 Homotopy and fixed endpoints

Write $I = [0, 1]$. Recall that a path in X is a continuous map $\gamma : I \to X$ (Lecture 2, connectedness and paths).

Definition 3.1 (Loop and base point)**.** A *loop* is a path with equal endpoints. Their common value is its *base point*.

Definition 3.2 (Homotopy)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $h : X \times I \to Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$ for every $x \in X$. We write $f \simeq g$. For paths from x_0 to x_1, a homotopy *relative to the endpoints* also satisfies

$$h(0, s) = x_0, \qquad h(1, s) = x_1 \qquad (s \in I).$$

Write $h_s(x) = h(x, s)$ for a fixed-parameter map in the homotopy. Continuity is required jointly in the path parameter t and deformation parameter s. For example, the paths $\gamma(t) = (t, 0)$ and $\delta(t) = (t, t(1 - t))$ in $\mathbb{R}^2$ are homotopic relative to their endpoints by

$$h(t, s) = (t, s\,t(1 - t)).$$

This map is continuous because its coordinates are polynomials. At $t = 0$ it is $(0, 0)$ and at $t = 1$ it is $(1, 0)$, independently of s. The figure shows the intermediate paths.

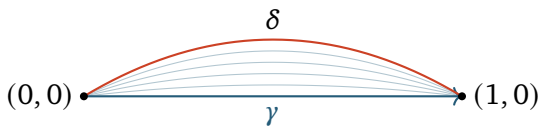


Figure 3.1: The homotopy keeps both endpoints fixed. Increasing s moves the straight path γ to the parabola δ.

We will glue formulas on finitely many closed pieces. If continuous maps on a finite closed cover agree on overlaps, their union is continuous. Indeed, the preimage of a closed set is a finite union of sets closed in those pieces, hence closed in the whole domain. This is the pasting argument of §2.3, now also applied to squares.

Homotopy relative to endpoints is an equivalence relation. The constant family proves reflexivity, and $h(t, 1 - s)$ reverses a homotopy. To join two homotopies, run the first on $0 \le s \le 1/2$ and the second on $1/2 \le s \le 1$, rescaling each parameter interval to I. They agree at the common edge, so the pasting argument proves transitivity.

The following loops illustrate how removing a point can obstruct a contraction. In the punctured plane, the circle once around the origin cannot contract. We prove this in §3.5. The circle centred at 2 with radius $2/5$ contracts by

$$h(t, s) = 2 + (1 - s)\frac{2}{5}e^{2\pi i t}, \qquad (t, s) \in I^2.$$

Its distance from 0 is at least $8/5$. A circle in three-space can instead contract through latitudes:

$$h(t, s) = \big(\cos(\pi s/2)\cos(2\pi t), \cos(\pi s/2)\sin(2\pi t), \sin(\pi s/2)\big).$$

Its norm is 1, and at $s = 1$ it is the constant north pole. This contracts this particular circle in $\mathbb{R}^3 \setminus \{0\}$. The figure shows these examples.

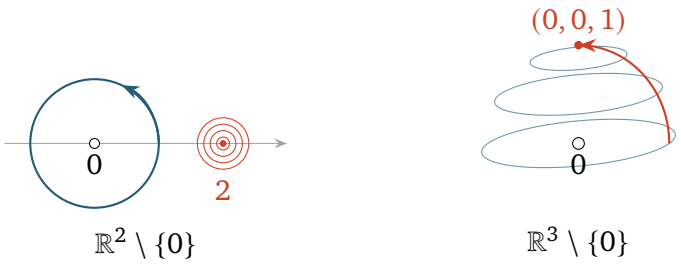


Figure 3.2: In the punctured plane, the circle around 0 and the circles centred at 2 behave differently under contraction. In three-space, the equatorial circle contracts to a pole without meeting 0.

Definition 3.3 (Free homotopy of loops). In a *free homotopy of loops* $h\colon I^2 \to X$, each h_s is a loop, but its base point may move. The defining endpoint condition is

$$h(0, s) = h(1, s) \qquad (s \in I).$$

3.2 Lifting phases on the circle

A circle point determines an angle only up to whole turns. We follow its angle continuously along a path, without resetting after a full turn. We measure this continuous phase in turns. Identify S^1 with the unit complex numbers, and set

$$p\colon \mathbb{R} \to S^1, \qquad p(u) = e^{2\pi i u}.$$

On I, this is the circle map c of Lecture 2, written with complex numbers. Here p denotes the exponential projection, rather than the stereographic projection of Lecture 1. We have $p(u) = p(v)$ exactly when $u - v \in \mathbb{Z}$.

Definition 3.4 (Lift and phase). For a continuous map $f\colon Y \to S^1$, a *lift* through p is a continuous map $\tilde{f}\colon Y \to \mathbb{R}$ with $p \circ \tilde{f} = f$. For this projection, a lift is also called a *phase*.

For example, the loop $\gamma(t) = p(2t)$ has lift $\tilde{\gamma}(t) = 2t$ starting at 0. It ends at 2, although the loop ends at 1 on the circle. Counter-clockwise travel $p(t)$ increases the phase by one turn. A principal phase in $(-1/2, 1/2]$ has its cut at -1 on the circle. For $p(2t)$ it is discontinuous at $t = 1/4$ and $t = 3/4$. It takes the value $+1/2$ at each cut time. The continuous lift has no such jumps.

The following figure compares these phases. The helix is the image of the map

$$H\colon \mathbb{R} \to \mathbb{R}^3, \qquad H(u) = (\cos(2\pi u), \sin(2\pi u), u),$$

whose projection onto the first two coordinates is $p(u)$. The loop $p(t + \sin(2\pi t))$ has lift $t + \sin(2\pi t)$. This lift takes the values $5/4$ at $t = 1/4$ and $-1/4$ at $t = 3/4$, but ends at 1.

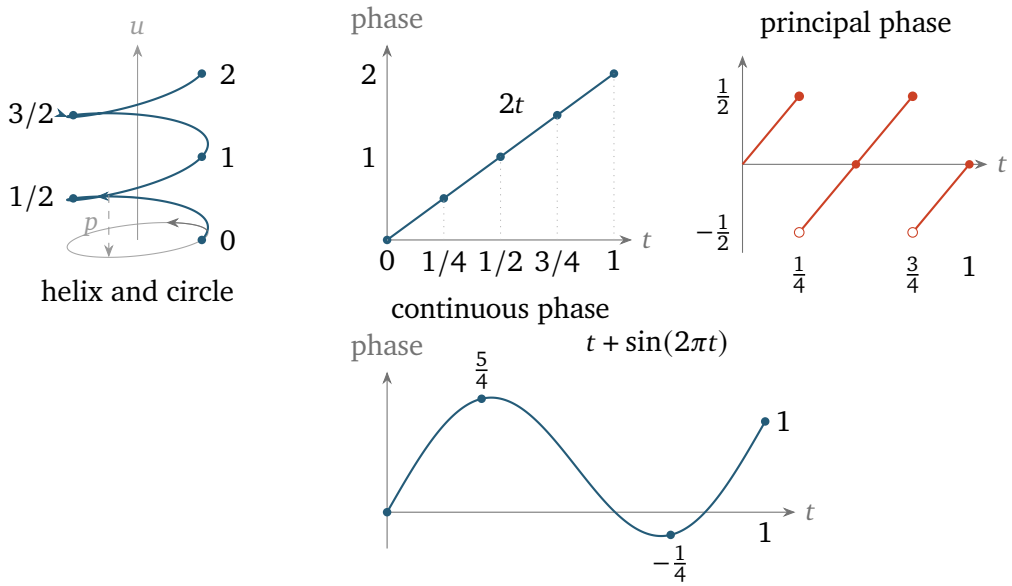


Figure 3.3: The loop $p(2t)$ makes two counter-clockwise turns: its continuous phase rises from 0 to 2, while its principal phase resets at the cut. The phase $t + \sin(2\pi t)$ reverses direction during the path and has total change 1.

For $r \in \mathbb{R}$ put

$$J_r = (r - 1/4, r + 1/4), \qquad U_r = \{z \in S^1 : \mathrm{Re}(p(-r)z) > 0\}.$$

The interval J_r has length $1/2$ and centre r. The set U_r is the open semicircle centred at $p(r)$. Multiplication by $p(-r)$ rotates it to the right half-circle U_0. These open semicircles cover the circle. The restriction $p\colon J_r \to U_r$ has the continuous inverse branch

$$b_0(z) = \frac{1}{2\pi} \arctan \frac{\operatorname{Im} z}{\operatorname{Re} z}, \qquad b_r(z) = r + b_0(p(-r)z).$$

For $z = e^{i\theta} \in U_0$ with $|\theta| < \pi/2$, the quotient $\operatorname{Im} z/\operatorname{Re} z$ is $\tan\theta$. The principal arctangent returns θ, so $b_0(z) = \theta/(2\pi) \in J_0$ and $p(b_0(z)) = z$. Conversely, for $u \in J_0$ the angle $2\pi u$ is in this range, so $b_0(p(u)) = u$. Rotating gives the inverse on U_r:

$$p(b_r(z)) = p(r)\,p(-r)z = z \qquad (z \in U_r).$$

The figure shows repeated real intervals for one semicircle.

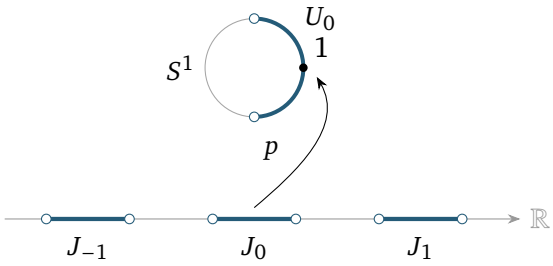


Figure 3.4: The intervals J_{-1}, J_0 and J_1 map to the same open semicircle U_0. Its boundary points are excluded.

We construct a lift by subdividing the path into pieces whose images lie in single semicircles. On each piece we use an inverse branch, then add an integer to match the preceding endpoint. For a homotopy we use the same construction on a square grid. Compactness of I and I^2 supplies a subdivision fine enough for every piece. The following lemma gives the uniform bound on their sizes. The *diameter* of a bounded nonempty set is the supremum of its pairwise distances.

Lemma 3.5 (Lebesgue number). *An open cover of a nonempty compact metric space K has a number $\delta > 0$ such that each nonempty subset of diameter less than δ lies in one cover member.*

Proof. For each $x \in K$ choose $\varepsilon_x > 0$ with $B_{2\varepsilon_x}(x)$ inside a cover member. Finitely many balls $B_{\varepsilon_x}(x)$ cover K. Let δ be the minimum of their radii. For a set A of diameter less than δ, choose $a \in A$ and one selected ball $B_{\varepsilon_x}(x)$ containing a. For every $z \in A$,

$$d(z,x) \le d(z,a) + d(a,x) < \delta + \varepsilon_x \le 2\varepsilon_x.$$

Thus A lies in the associated cover member. ∎

Integer principle. A continuous integer-valued function on a connected space is constant. Indeed, if two different integers occurred, a noninteger threshold between them would separate the domain into two nonempty open inverse images. We apply this principle to differences of lifts and then to their endpoint differences.

Lemma 3.6 (Path and homotopy lifting for the circle). *A path $\gamma\colon I \to S^1$ has a unique lift after specifying $\tilde\gamma(0) \in p^{-1}(\{\gamma(0)\})$. A homotopy $h\colon I^2 \to S^1$ has a unique lift after a lift of its bottom edge $h(t,0)$ has been specified.*

Proof. The interval and square are compact by Heine–Borel. Apply Lemma 3.5 to the cover of I by $\gamma^{-1}(U_r)$. For its Lebesgue number δ, choose N with $1/N < \delta$ and put $t_i = i/N$ for $i = 0, \dots, N$. Each subinterval maps into some U_r. On $[t_i, t_{i+1}]$, set $\tilde{\gamma} = b_r \circ \gamma + k$, with

$$k = \tilde{\gamma}(t_i) - b_r(\gamma(t_i)) \in \mathbb{Z}.$$

This matches the value already assigned at t_i. The pieces agree at their endpoints and paste continuously.

For a homotopy, use the cover of I^2 by $h^{-1}(U_r)$. Choose a square grid whose cells have diameter less than its Lebesgue number. Extend the given bottom lift across cells, row by row, from left to right. On a cell choose $b_r \circ h + k$ to match the assigned value at its lower-left corner. The only previously assigned points of a new cell are on its bottom and left edges. Along each already assigned bottom or left edge, the two lifts differ by a continuous integer-valued function. That function is constant by connectedness, and zero at the corner. Thus the new piece agrees on shared edges and pastes continuously.

Finally, two lifts on I or I^2 differ by a continuous integer-valued function. The integer principle makes that difference constant. Agreement at one point therefore proves uniqueness. ∎

The figure shows one extension step of the grid construction.

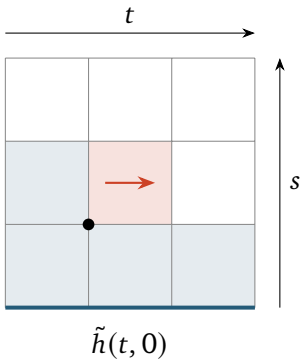


Figure 3.5: Previously assigned cells are shaded. A branch on the next cell is shifted by an integer to agree at the marked corner and hence on the shared edges.

Definition 3.7 (Winding number)**.** For any loop γ in the circle, define its *winding number* by

$$\operatorname{wind} \gamma = \tilde{\gamma}(1) - \tilde{\gamma}(0) \in \mathbb{Z}.$$

Different initial lifts differ by an integer, so this number is independent of that choice. In a free homotopy of loops, lift the whole square. The endpoint difference is a continuous integer-valued function of s, so it is constant. Thus winding is invariant even when the base point moves.

For $p(2t)$ the endpoint difference is 2. For $p(t + \sin(2\pi t))$ it is 1, despite the changes of direction. A constant loop has a constant lift and winding 0. Hence neither of these two loops is freely homotopic to a constant loop.

Self-check. Find the winding number of the circle loop

$$\gamma(t) = \frac{3/5 + p(2t)}{|3/5 + p(2t)|}\, p(-5t), \qquad t \in I.$$

Construct a free homotopy to a loop with an explicit lift. Justify that its normalization is defined throughout, and decide whether γ contracts in the circle.

Answer. We use the free homotopy

$$h(t,s) = \frac{3s/5 + p(2t)}{|3s/5 + p(2t)|}\, p(-5t).$$

The numerator in the fraction has modulus at least $1 - 3s/5 \geq 2/5$. Thus the formula is continuous, and its endpoints in t agree for every $s \in I$. At $s = 0$ it is $p(2t)p(-5t) = p(-3t)$, with lift $-3t$. At $s = 1$ it is γ. Free-homotopy invariance gives $\operatorname{wind}\gamma = -3$. Since a constant loop has winding 0, the loop cannot contract in the circle.

Unit 3B Loop classes and the fundamental group

3.3 The fundamental group

For paths γ from x to y and δ from y to z, define

$$(\gamma \cdot \delta)(t) = \begin{cases} \gamma(2t), & 0 \leq t \leq \frac{1}{2}, \\ \delta(2t-1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The convention is γ **first, then** δ, as in Hatcher, §1.1. Define the reversed path by $\bar{\gamma}(t) = \gamma(1-t)$, and the constant path by $e_x(t) = x$. All these paths are continuous, by pasting at the matching endpoints. Write $[\gamma]$ for the endpoint-fixed homotopy class of γ.

Theorem 3.8 (Fundamental group). *The endpoint-fixed homotopy classes of loops at $x_0 \in X$ form a group, written $\pi_1(X, x_0)$, with*

$$[\gamma][\delta] = [\gamma \cdot \delta], \qquad 1 = [e_{x_0}], \qquad [\gamma]^{-1} = [\bar{\gamma}].$$

Proof. We verify the group laws using endpoint-fixed homotopies. Concatenation of paths is not literally associative: the bracketings allot different times to the three paths. We make their classes associative by changing those times continuously.

Products respect classes. Concatenate two endpoint-fixed homotopies in the t direction. Their formulas agree along $t = 1/2$. Reversing the t parameter similarly respects classes.

A change of timing does not change a path class. For any continuous $\rho\colon I \to I$ with $\rho(0) = 0$ and $\rho(1) = 1$, the formula $h(t,s) = \gamma((1-s)\rho(t) + st)$ joins $\gamma \circ \rho$ to γ while fixing the endpoints. For associativity, the following timing changes $((\gamma \cdot \delta) \cdot \eta) \circ \rho$ into $\gamma \cdot (\delta \cdot \eta)$:

$$\rho(t) = \begin{cases} t/2, & 0 \leq t \leq \frac{1}{2}, \\ t - \frac{1}{4}, & \frac{1}{2} \leq t \leq \frac{3}{4}, \\ 2t - 1, & \frac{3}{4} \leq t \leq 1. \end{cases}$$

The formulas agree at the break times and fix 0 and 1. On the first interval the parameter in γ becomes $4\rho(t) = 2t$. On the middle interval it becomes $4\rho(t) - 1 = 4t - 2$ in δ. On the

last it becomes $2\rho(t) - 1 = 4t - 3$ in η. These are precisely the timings of $\gamma \cdot (\delta \cdot \eta)$. The preceding homotopy therefore proves associativity of classes.

The paths $e_x \cdot \gamma$ and $\gamma \cdot e_y$ are γ with timings $\max(2t-1, 0)$ and $\min(2t, 1)$. Thus their classes equal $[\gamma]$. Finally, the homotopy

$$h(t,s) = \gamma\big((1-s)\min(2t, 2-2t)\big)$$

contracts $\gamma \cdot \bar{\gamma}$ to e_x, fixing both endpoints. At deformation time s, it follows γ only as far as parameter $1-s$ and then returns along the same part of γ. Applying it to $\bar{\gamma}$ gives the other inverse identity. These homotopies also apply to joined paths with different endpoints. ∎

The figure records the two timings and the shortening of the return path.

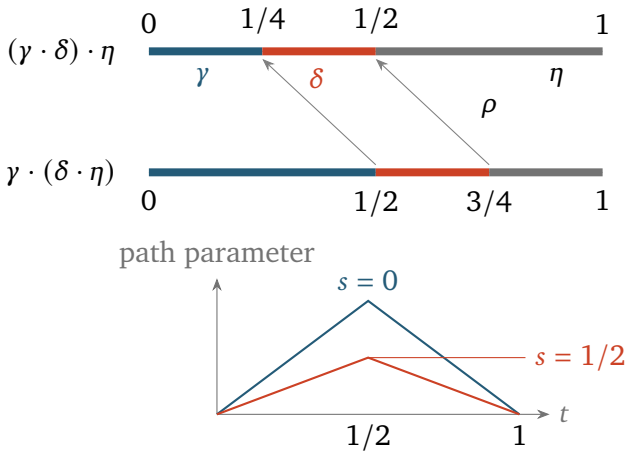


Figure 3.6: The map ρ aligns the break times of the two bracketings. In the return-path homotopy, the path parameter rises only to $1-s$ and then falls to 0.

Definition 3.9 (Homomorphism and isomorphism)**.** A *homomorphism* preserves group products. A bijective homomorphism is an *isomorphism*.

A continuous map $f\colon X \to Y$ induces

$$f_*\colon \pi_1(X, x_0) \longrightarrow \pi_1(Y, f(x_0)), \qquad [\gamma] \longmapsto [f \circ \gamma].$$

Proposition 3.10 (Induced homomorphisms)**.** *The induced map f_* is a homomorphism. Induced maps preserve composition and identity maps.*

Proof. Composing a path homotopy with f proves well-definedness. Composition also respects the two path pieces, so f_* preserves products. For continuous $g\colon Y \to Z$, this gives $(g \circ f)_* = g_* \circ f_*$. Identity maps give $(\mathrm{id}_X)_* = \mathrm{id}$. ∎

Theorem 3.11 (Fundamental group of the circle)**.** *Winding gives an isomorphism* $\mathrm{wind}\colon \pi_1(S^1, 1) \to \mathbb{Z}$. *The loop $t \mapsto p(mt)$ has winding m for every $m \in \mathbb{Z}$.*

Proof. Winding is invariant under free homotopy, hence under endpoint-fixed homotopy. It therefore defines a map on classes. Choose a lift starting at 0. If it ends at m, the homotopy

$$\tilde{h}(t,s) = (1-s)\tilde{\gamma}(t) + smt$$

fixes the endpoints 0 and m. Its projection gives an endpoint-fixed homotopy from γ to $p(mt)$. Thus winding determines the class, and every integer occurs. For a product of loops

with windings m and n, join the first lift to the second lift shifted by m. The joined lift ends at $m + n$, so winding preserves multiplication as addition. ∎

The translated second lift in the proof is shown for windings 1 and 2. Its final height is 3, the sum of the two endpoint changes.

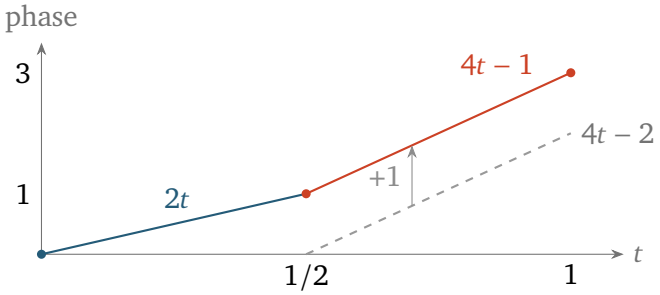


Figure 3.7: For $p(t) \cdot p(2t)$, the second lift starts at height 1. The joined lift is $2t$ on the first half and $4t - 1$ on the second half.

3.4 Base points and free homotopy

The base point requires care. For a path a from x_0 to x_1, the base-point change map is

$$c_a \colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad c_a([\gamma]) = [\bar{a} \cdot \gamma \cdot a].$$

The figure gives the traversal order.

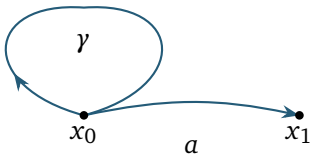


Figure 3.8: A loop at x_1 follows $\bar{a}$ to x_0, traverses γ, then follows a back.

Proposition 3.12 (Change of base point)**.** *For every path a from x_0 to x_1, the map c_a is an isomorphism.*

Proof. Concatenating with fixed paths respects homotopy, so c_a is well-defined. Cancellation of $a \cdot \bar{a}$ proves that c_a preserves products. The inverse is $c_{\bar{a}}$, by the same cancellation. Thus it is an isomorphism. ∎

Definition 3.13 (Conjugate elements)**.** Elements u and v of a group are *conjugate* if $v = k^{-1}uk$ for some element k of that group.

When $x_0 = x_1$, the map c_a is conjugation by $[a]$. If $[a][\gamma] \neq [\gamma][a]$, then $c_a([\gamma]) \neq [\gamma]$, whereas the constant choice of a gives the identity map. Thus the isomorphism can depend on the chosen path.

For example, $h(t, s) = p(t + s/4)$ is a free homotopy from $p(t)$ to $p(t + 1/4)$. Its base point follows $a(s) = p(s/4)$ from 1 to i. Its continuous phase $t + s/4$ has endpoint difference 1 for every $s \in I$.

We now convert a free homotopy into a homotopy with a fixed base point. Let h be a free homotopy of loops. Let $\gamma_0 = h_0$ be based at x_0, and put $\gamma_1 = h_1$. The base point follows the path $a(s) = h(0, s)$. Put $a_s(t) = a(st)$. The loops $a_s \cdot h_s \cdot \bar{a}_s$ keep base point x_0. Each first follows a_s from x_0, then traverses h_s, then returns along $\bar{a}_s$. At $s = 0$, the inserted constant paths do not change the class, by the unit laws. At $s = 1$ this gives $[\gamma_0] = [a \cdot \gamma_1 \cdot \bar{a}]$. If γ_1 is also based at x_0, then a is a loop at x_0. The equality therefore makes $[\gamma_0]$ and $[\gamma_1]$ conjugate.

Conversely, let a be a loop at x_0. The family $\bar{a}_s \cdot \gamma \cdot a_s$ consists of loops with base point $a(s)$. Its end loops represent $[\gamma]$ and $[\bar{a} \cdot \gamma \cdot a]$. For each family, the three formulas are jointly continuous in t and s. They agree on the closed boundaries between their parameter pieces, so pasting proves joint continuity. Free homotopy is an equivalence relation by the same pasting argument as before. Based homotopies are free homotopies, so conjugate classes are freely homotopic. Thus two loops based at x_0 are freely homotopic exactly when their classes are conjugate. In a path-connected space, move any base point to x_0 by a path. Free homotopy classes then correspond to conjugacy classes in $\pi_1(X, x_0)$, rather than to individual group elements.

The parameter cylinder in the figure records the moving base point and the inserted paths. Its image under h need not be an embedded annulus in X.

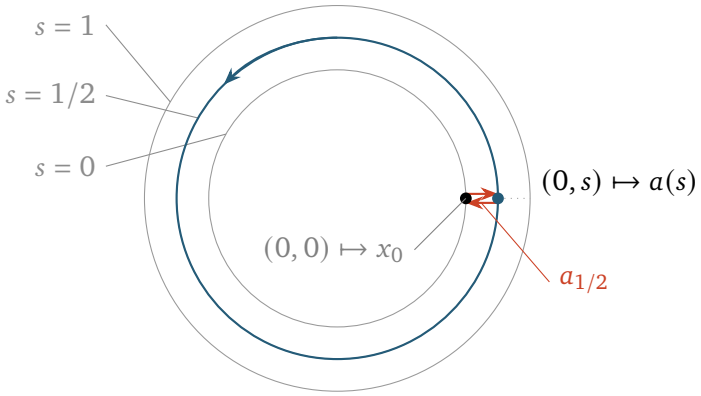


Figure 3.9: The annulus represents the parameter cylinder by $(t, s) \mapsto (1 + s/2)p(t)$. The marked radial segment represents $a_{1/2}$, followed outward and then back in the based construction.

For the circle group, conjugation changes no class, since addition in $\mathbb{Z}$ commutes. Thus two loops based at 1 are freely homotopic exactly when they have the same winding. In a noncommutative group, $k^{-1}uk$ can differ from u. Lecture 5 computes such fundamental groups using reduced words.

3.5 Retractions and homotopy invariance

Definition 3.14 (Retraction and deformation retraction)**.** A *retraction* onto $A \subseteq X$ is a continuous map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$. A *deformation retraction* also has a homotopy from id_X to $i \circ r$, where $i\colon A \to X$ is inclusion, that keeps every point of A fixed throughout.

Proposition 3.15 (Deformation retractions preserve the group)**.** *For a deformation retraction and $a_0 \in A$, the induced maps r_* and i_* are inverse isomorphisms of fundamental groups.*

Proof. We have $r \circ i = \mathrm{id}_A$. The other composite is homotopic to id_X while fixing a_0. Composing this homotopy with any loop at a_0 is an endpoint-fixed path homotopy. Hence $i_* r_* = \mathrm{id}$ as well as $r_* i_* = \mathrm{id}$. ∎

Identify $\mathbb{C}$ with $\mathbb{R}^2$. For example, on $\mathbb{C} \setminus \{0\}$ the formula

$$h(z, s) = \left(1 - s + \frac{s}{|z|}\right) z$$

deforms the identity to $r(z) = z/|z|$. Its coefficient is positive, so the image never reaches 0. The homotopy fixes every point of S^1. Consequently $\pi_1(\mathbb{R}^2 \setminus \{0\}, 1) \cong \mathbb{Z}$. For a loop in the punctured plane, its winding means the winding of its normalization $\gamma/|\gamma|$. The loop $p(t)$ has winding +1, whereas a constant loop has winding 0. Therefore it cannot contract in the punctured plane. This proves the obstruction pictured in Unit 3A.

Definition 3.16 (Simply connected space)**.** A nonempty space is *simply connected* if it is path-connected and all based loops are endpoint-fixed homotopic to the constant loop.

Definition 3.17 (Contractible space)**.** A space is *contractible* if its identity is homotopic to a constant map with value x_* in the space.

A nonempty convex subset of $\mathbb{R}^n$ is contractible: for a chosen point x_*, the straight-line homotopy $h(x, s) = (1 - s)x + s x_*$ stays in the subset.

Proposition 3.18 (Contractible spaces)**.** *Every contractible space is simply connected.*

Proof. The contraction gives paths from all points to x_*, so the space is path-connected. For a loop γ at x_0, it also gives a free homotopy to e_{x_*}. The moving-basepoint construction above converts this to a based homotopy from γ to $a \cdot e_{x_*} \cdot \bar{a}$, which cancels to e_{x_0}. ∎

The punctured plane is not contractible, since its fundamental group is $\mathbb{Z}$.

Definition 3.19 (Homotopy equivalence)**.** Maps $f\colon X \to Y$ and $g\colon Y \to X$ give a *homotopy equivalence* if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$.

The punctured plane and the circle are homotopy equivalent by the radial retraction and inclusion. Their two composites are the identity on the circle and a map homotopic to the identity on the punctured plane, as the radial homotopy proves. The general theorem that such f induces an isomorphism on fundamental groups at every base point is cited from Hatcher, Proposition 1.18. The deformation-retraction case used here was proved directly above.

3.6 No retraction of the disc onto its circle

A retraction would assign a boundary point continuously to every point of the disc while fixing each boundary point. The obstruction is the loop that contracts through the disc but cannot contract along its boundary. Let $D^2 = \{z \in \mathbb{C} : |z| \le 1\}$ be the closed disc.

Corollary 3.20. *There is no retraction $r\colon D^2 \to S^1$.*

Proof. The homotopy $(z, s) \mapsto (1 - s)z$ contracts the disc. Thus $\pi_1(D^2, 1)$ is the trivial group, consisting only of its identity. If r were a retraction and $i\colon S^1 \to D^2$ the inclusion, then $r \circ i = \mathrm{id}_{S^1}$. Therefore $r_* i_*$ is the identity of $\pi_1(S^1, 1) \cong \mathbb{Z}$. But i_* takes values in the trivial group, so the composite sends every class to the identity element. It cannot be the identity map of this nontrivial group. ∎

The base point here is 1 on the circle. Contractibility gives the trivial group there even though the displayed contraction ends at the centre.

The corollary gives the fixed-point theorem in the plane.

Corollary 3.21 (Brouwer's theorem in the plane)**.** *Every continuous map $f\colon D^2 \to D^2$ has a fixed point.*

Proof. We argue by contradiction. Suppose that $f(z) \neq z$ for every $z \in D^2$, and put $u(z) = z - f(z)$, so that $u(z) \neq 0$. We follow the ray from $f(z)$ through z until it meets the circle. The point $z + tu(z)$ with $t \geq 0$ lies on the circle exactly when

$$|u(z)|^2 t^2 + 2\,\mathrm{Re}\big(z\overline{u(z)}\big)t + |z|^2 - 1 = 0.$$

The product of the two roots is $(|z|^2 - 1)/|u(z)|^2 \leq 0$, so the larger root is nonnegative. We take $t(z)$ to be this root:

$$t(z) = \frac{-\,\mathrm{Re}\big(z\overline{u(z)}\big) + \sqrt{\mathrm{Re}\big(z\overline{u(z)}\big)^2 + |u(z)|^2(1 - |z|^2)}}{|u(z)|^2}.$$

This formula is continuous in z, because $u(z) \neq 0$ and the quantity under the root is nonnegative. For $|z| = 1$ we have $\mathrm{Re}\big(z\overline{u(z)}\big) = 1 - \mathrm{Re}\big(z\overline{f(z)}\big) \geq 1 - |f(z)| \geq 0$, so $t(z) = 0$. Hence $r(z) = z + t(z)u(z)$ is a retraction of D^2 onto S^1, which contradicts Corollary 3.20. ∎

Lecture 14 proves the same theorem for smooth maps of balls in every dimension.

Self-check. State in one sentence what the fundamental group of (X, x_0) is and why we introduced it. Explain why paths themselves are replaced by classes.

Answer. The fundamental group is the group of endpoint-fixed homotopy classes of loops at x_0, introduced to distinguish loops that cannot deform into one another while retaining concatenation. The two bracketings of three paths have different timings, but a change of timing is an endpoint-fixed homotopy. Thus associativity holds for classes.

Self-check. Compute $\pi_1(S^1 \times S^1, (1, 1))$ using the coordinate projections. Prove the group identification, not just its surjectivity.

Answer. Send $[(\gamma, \delta)]$ to $(\mathrm{wind}\,\gamma, \mathrm{wind}\,\delta) \in \mathbb{Z}^2$. Projecting a product homotopy gives endpoint-fixed homotopies of the coordinates. Winding is invariant under these, so the map is well-defined. Projection respects concatenation, so the map preserves multiplication coordinatewise. Conversely, the component test of Lecture 1 makes a pair of loops a loop in the product, and a pair of endpoint-fixed homotopies a homotopy in the product. For every $(m, n) \in \mathbb{Z}^2$, the pair is represented by $t \mapsto (p(mt), p(nt))$. If both windings vanish, each coordinate loop has an endpoint-fixed contraction. Their pair contracts the product loop. If two classes have the same image, their product with the inverse of the second has both windings zero, so is trivial. The two classes therefore agree. The group is $\mathbb{Z}^2$.

Unit 3C Polynomial roots and winding

3.7 The fundamental theorem of algebra

Corollary 3.22 (Fundamental theorem of algebra)**.** *Every complex polynomial of degree at least one has a complex root.*

Proof. We compare the same large-circle loop in two ways. On a sufficiently large circle the leading term dominates, so removing the lower terms without meeting zero gives winding equal to the degree. If the polynomial had no root anywhere, shrinking the input circle to its centre would also avoid zero and give winding zero. We prove these two claims.

Write $f(z) = a_n z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$ and $a_n \neq 0$. Choose

$$R > \max\left\{1, \frac{\sum_{j=0}^{n-1} |a_j|}{|a_n|}\right\}.$$

The diagram separates the two deformations. The second row assumes that f has no zero anywhere in $\mathbb{C}$.

$$\begin{array}{lccc} \text{large circle} & a_n R^n p(nt) & \xrightarrow{\text{add lower terms}} & f(Rp(t)) \\ & \text{winding } n & & \text{winding } n \\ \\ \text{no-root assumption} & f(Rp(t)) & \xrightarrow{\text{shrink input circle}} & f(0) \\ & \text{winding } n & & \text{winding } 0 \end{array}$$

Figure 3.10: The large-circle comparison keeps winding n while adding the lower terms. Under the no-root assumption, shrinking the input circle would instead give winding 0 for the same loop.

The large circle. For $|z| = R$ the lower terms satisfy

$$\sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < |a_n| R^n.$$

Thus the loops

$$a_n R^n p(nt) + s \sum_{j=0}^{n-1} a_j R^j p(jt)$$

avoid zero for every $s \in I$. Normalizing gives a free homotopy of circle loops from the leading term to $f(Rp(t))$. Choose $b \in \mathbb{R}$ with $p(b) = a_n/|a_n|$. The leading loop has phase $b + nt$, so its winding is n.

The figure illustrates this comparison for a quadratic.

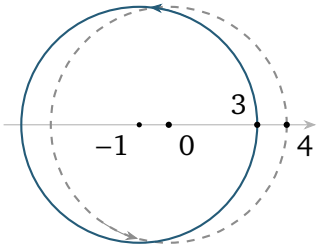


Figure 3.11: For $f(z) = z^2 - 1$ and $R = 2$, the comparison loops are $4p(2t) - s$, each traversed twice. The dashed loop ($s = 0$) and solid loop ($s = 1$) start at 4 and 3 and have centres 0 and −1, respectively.

The shrinking circle. Suppose f had no zero anywhere in $\mathbb{C}$. The map $h_f \colon I^2 \to \mathbb{C} \setminus \{0\}$ given by

$$h_f(t, s) = f((1 - s)Rp(t))$$

would be a homotopy of nonzero loops from $f(Rp(t))$ to the constant $f(0)$. These families may move the base point. After normalization, free-homotopy invariance would preserve winding and give its final value zero. This contradicts its value $n \geq 1$. Hence f has a root. ∎

The quadratic $f(z) = z^2 - 1$ shows where a shrinking family can meet zero. For an input circle of radius ρ, its image is $\rho^2 p(2t) - 1$. At $\rho = 1$ it meets 0 at $t = 0$ and $t = 1/2$, corresponding to the roots 1 and −1. The figure compares these input and image circles.

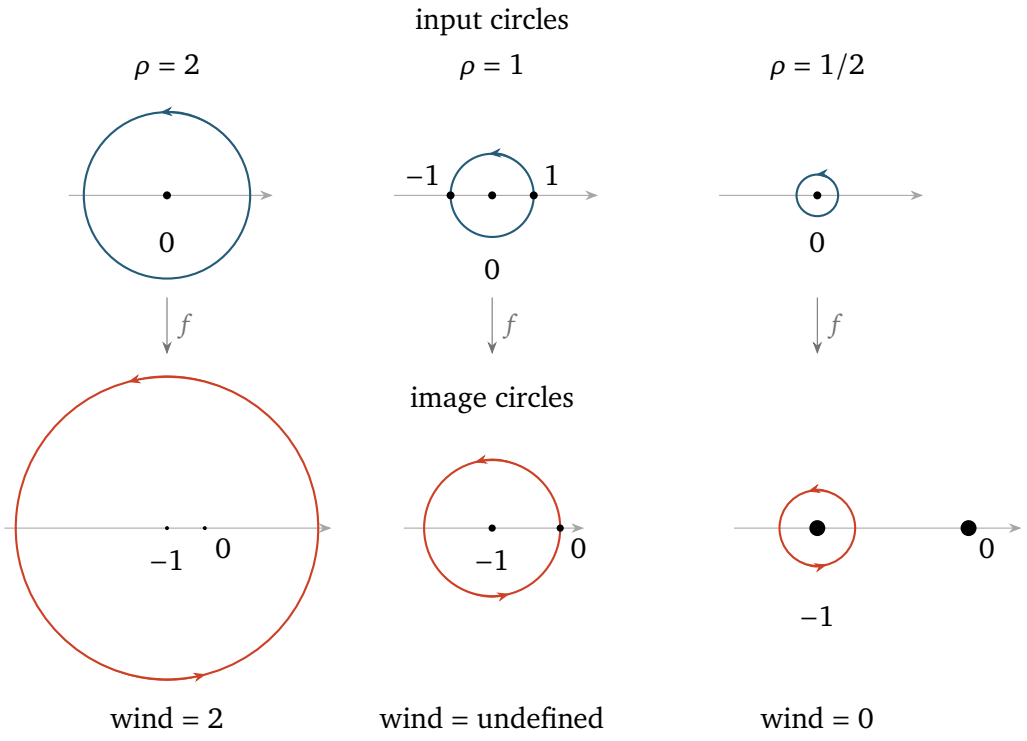


Figure 3.12: For input radii 2, 1 and 1/2, the image circles, drawn at different scales, have radii 4, 1 and 1/4 about −1 and are each traversed twice. The middle image meets 0, so its winding about 0 is undefined.

Summary. Classes of loops at one fixed base point form the fundamental group. Changing the base point along a path gives an isomorphism that can depend on that path. In a

path-connected space, free homotopy keeps only conjugacy classes. Lifting computes $\pi_1(S^1, 1) \cong \mathbb{Z}$. Deformation retraction gives the same group for the punctured plane. This integer rules out a disc-to-circle retraction and proves the fundamental theorem of algebra. Lecture 4 extends the lifting construction to other maps.

Reading. Hatcher, *Algebraic Topology*, §1.1, pp. 25–38 (Theorem 1.8 and Proposition 1.18); Munkres, *Topology*, §§51–56, §58 and Lemma 27.5.